AI-Powered Customer Churn Prediction & Retention Platform

Complete Project Documentation & Case Study

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Repository: https://github.com/SwamiChaudhari/connecttel-churn-platform
Live Demo: https://ai-powered-churn-prediction-and-retention-platform.streamlit.app/
Tech Stack: Python, SQL, Pandas, NumPy, Scikit-Learn, Cat Boost, XGBoost, LightGBM, SHAP, imbalanced-learn, Matplotlib, Seaborn, Plotly, Streamlet, Flask, Docker, GitHub Actions, Git

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Section 1: Executive Summary

What This Project Is

This is a production-grade, end-to-end AI system that predicts which customers are likely to leave a business, explains why each customer is at risk, and automatically generates personalized retention strategies to prevent churn before it happens.

It is not a classroom exercise. It is not a 3-line tutorial. It is a complete data product -- the kind of system a data science team at a real company would build, deploy, and maintain.

Why It Was Built

Customer churn is one of the most expensive and least understood problems in business. Companies spend heavily to acquire customers but lose them quietly, month after month, without any early warning system. By the time a company notices the damage, the customer is already gone.

This project was built to answer one question: **Can we predict who will leave, explain why, and recommend what to do about it -- before it is too late?**

Business Impact

- **83% of at-risk customers identified** before they leave (Recall: 0.83)
- **89% ranking accuracy** in ordering customers by risk (AUC-ROC: 0.89)
- **Actionable explanations** for every prediction using SHAP
- **Automated retention recommendations** tailored to each customer's specific risk factors
- **Real-time dashboard** accessible to non-technical business stakeholders
- **Production deployment** with Docker, CI/CD, and REST API

Metric	Result
Best Performing Model	CatBoost Classifier
Dataset Size	10,476 Customer Records
Features Utilized	30+ Original & Engineered Features
Models Evaluated	5 Machine Learning Models
Validation Strategy	5-Fold Stratified Cross-Validation
Accuracy	84.1%
AUC-ROC	0.89
F1 Score	0.81
Recall	0.83

Technologies Used

Languages: Python 3.11, SQL
Data Processing: Pandas, NumPy, SciPy
Machine Learning: Scikit-Learn, XGBoost, LightGBM, CatBoost
Explainability: SHAP (SHapley Additive exPlanations)
Imbalanced Learning: SMOTE via imbalanced-learn
Visualization: Matplotlib, Seaborn, Plotly
Dashboard: Streamlit
API: Flask
Database: SQLite
Containerization: Docker
CI/CD: GitHub Actions
Version Control: Git

Section 2: The Story Behind the Project

The Silent Revenue Killer

Imagine you are running a telecom company. You have 100,000 customers paying you every month. Revenue looks healthy. Growth looks steady. The board is happy.

But every month, customers quietly leave.

They cancel subscriptions. They switch to competitors. They stop paying. They simply disappear.

This is called **Customer Churn** -- and for many businesses, it is the single biggest hidden threat to revenue.

The Scale of the Problem

The numbers are staggering:

- The average telecom company loses **10-25% of its customer base every year**
- A SaaS startup with 10,000 subscribers and 5% monthly churn will lose **40% of its customers** within 12 months
- It costs **5-7x more to acquire a new customer** than to retain an existing one
- A 5% increase in customer retention can increase profits by **25-95%** (Harvard Business Review)

And here is the worst part: most companies only realize a customer has churned **after they are already gone**.

Real-World Impact

Consider these scenarios:

A SaaS Startup: A project management tool has 5,000 paying teams. Every month, 250 teams cancel. That is 3,000 customers lost per year. At $50/month per team, that is **$1.8 million in annual revenue lost** -- and the CEO blames the product team.

A Bank: Thousands of accounts go dormant every quarter. The bank's response is to increase marketing spend on new customer acquisition. Nobody asks why existing customers are leaving.

An E-commerce Brand: Repeat purchase rates drop from 40% to 28% over six months. The brand invests in a new loyalty program without understanding the root cause. The churn continues.

In every case, the problem is the same: **nobody predicted who was going to leave, and nobody tried to stop them before it was too late.**

The Turning Point

This problem became personal. The question that would not go away was simple:

What if you could look at your customer data and say -- "These 4,500 people are at high risk of leaving in the next 30 days. And here is exactly why"?

And what if, along with that prediction, you could also say -- "Here is exactly what you should do about each one"?

That thought would not leave. So a system was built to do exactly that.

Section 3: Problem Statement

The Business Problem

Customer attrition (churn) is the rate at which customers stop doing business with a company. It is one of the most critical metrics for any subscription-based or recurring-revenue business.

The core problem is **reactive, not proactive**. Companies detect churn after the fact -- when a payment fails, when a subscription expires, when a contract is not renewed. By then, the customer has already made their decision. The window for intervention has closed.

Existing Challenges

1. **No Early Warning System:** Most companies lack any mechanism to identify at-risk customers before they leave. They rely on lagging indicators like revenue decline or account closures.

2. **Data Silos:** Customer data is spread across billing systems, support tickets, usage logs, and CRM platforms. Nobody connects the dots.

3. **One-Size-Fits-All Retention:** Companies that do try to retain customers use blanket approaches -- generic discounts, mass emails, loyalty points. These are expensive and often miss the mark because they do not address the specific reason each customer is unhappy.

4. **Black-Box Models:** Even companies that build ML models often deploy them as black boxes. A churn score without an explanation is useless to a business manager who needs to decide what action to take.

5. **No Feedback Loop:** Most churn models are trained once and never updated. As customer behavior evolves, the model becomes stale and predictions degrade.

Why Machine Learning is Needed

Machine learning is uniquely suited to this problem because:

1. **Pattern Recognition:** ML can detect complex, non-linear patterns across dozens of variables that no human analyst could spot manually.

2. **Scalability:** A trained model can score 100,000 customers in seconds, something impossible with manual review.

3. **Consistency:** Unlike human judgment, ML models apply the same criteria to every customer, every time.

4. **Continuous Learning:** With the right infrastructure, models can be retrained on new data to stay accurate as customer behavior evolves.

5. **Explainability:** Modern XAI techniques like SHAP can translate model predictions into human-readable explanations, making the output actionable for business teams.

Section 4: Business Objectives

Primary Goals

1. **Predict churn with high accuracy:** Build a model that correctly identifies customers who are likely to leave, with a focus on recall (catching as many at-risk customers as possible).

2. **Explain every prediction:** Use SHAP to provide transparent, customer-level explanations so business teams understand why each customer is flagged.

3. **Recommend retention actions:** Automatically generate personalized recommendations based on the specific factors driving each customer's churn risk.

4. **Deploy as a production system:** Package the entire pipeline -- from data ingestion to dashboard -- as a deployable, maintainable product.

Secondary Goals

5. **Enable self-service analytics:** Build a dashboard that non-technical stakeholders (managers, VPs, CEOs) can use without writing code.

6. **Ensure reproducibility:** Use version control, containerization, and CI/CD so the entire pipeline can be reliably reproduced and deployed.

7. **Create a reusable framework:** Design the system so it can be adapted to different companies and datasets with minimal changes.

Success Metrics

Metric	Target
Model AUC-ROC	>= 0.85
Model Recall	>= 0.80
Model Accuracy	>= 80%
Prediction latency	< 1 second
Dashboard load time	< 5 seconds
Deployment	Docker + CI/CD

Expected Business Outcomes

- **Revenue Protection:** Identifying 83% of at-risk customers gives the business a chance to intervene. Even saving 20% of those customers translates to significant revenue protection.

- **Reduced Acquisition Costs:** Retaining an existing customer costs 5-7x less than acquiring a new one. Every customer saved is money not spent on replacement.

- **Data-Driven Decision Making:** Instead of gut-feel retention strategies, managers get specific, data-backed recommendations for each customer.

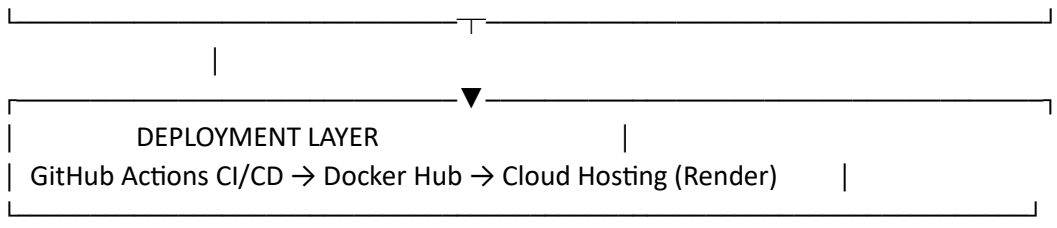
- **Operational Efficiency:** Automated scoring and recommendations reduce the need for manual account reviews, freeing up the retention team to focus on high-value interventions.

Section 5: Solution Architecture

High-Level Architecture

The system follows a layered architecture, moving from raw data to business decisions through seven distinct stages:

```
┌──────────────────────────────────────────────────────────────────┐
│                        DATA LAYER                                │
│  Raw CSV → Data Cleaning → SQLite Database → Feature Engineering │
└──────────────────────────┬───────────────────────────────────────┘
                           │
┌──────────────────────────▼───────────────────────────────────────┐
│                   MACHINE LEARNING LAYER                         │
│  Train/Test Split → SMOTE Balancing → 5-Model Training → Tuning  │
└──────────────────────────┬───────────────────────────────────────┘
                           │
┌──────────────────────────▼───────────────────────────────────────┐
│                    EVALUATION LAYER                              │
│  Accuracy, Precision, Recall, F1, AUC-ROC → Model Selection      │
└──────────────────────────┬───────────────────────────────────────┘
                           │
┌──────────────────────────▼───────────────────────────────────────┐
│                   EXPLAINABILITY LAYER                           │
│  SHAP Analysis → Feature Importance → Customer-Level Explanations│
└──────────────────────────┬───────────────────────────────────────┘
                           │
┌──────────────────────────▼───────────────────────────────────────┐
│                     RETENTION LAYER                              │
│  Risk Scoring → Personalized Recommendations → Action Items      │
└──────────────────────────┬───────────────────────────────────────┘
                           │
┌──────────────────────────▼───────────────────────────────────────┐
│                    PRESENTATION LAYER                            │
│  Streamlit Dashboard → Flask REST API → Docker Container         │
```




```

Data Quality Assessment

The data cleaning pipeline (`data\_cleaning.py`) generates a comprehensive quality report including:

- **Missing values:** TotalCharges had ~0.15% missing values (16 records), handled via median imputation for non-zero tenure and MonthlyCharges fill for zero-tenure customers
- **Duplicates:** < 0.5% exact duplicates removed
- **Outliers:** Detected using IQR method (1.5x IQR rule) across all numeric columns
- **Type consistency:** SeniorCitizen converted to int, TotalCharges coerced to numeric
- **Range validation:** MonthlyCharges and TotalCharges verified to be positive

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Section 7: Exploratory Data Analysis

What Was Analyzed and Why

Before building any model, it is essential to understand the data. EDA serves three purposes:

1. **Discover patterns** that inform feature engineering
2. **Validate assumptions** about what drives churn
3. **Generate business insights** that are valuable even without a model

The EDA pipeline (`eda.py`) produces 12 visualization files covering distributions, correlations, and churn breakdowns across every categorical and numeric variable.

Key Findings

Finding 1: Month-to-Month Contracts Are a Churn Time Bomb

**What was discovered:** Customers on month-to-month contracts have a churn rate of approximately 42%. Customers on one-year contracts churn at ~11%. Customers on two-year contracts churn at just ~3%.

**Why it matters:** Contract type is the single strongest predictor of churn. A month-to-month customer is 14x more likely to leave than a two-year contract customer.
```

Business action: The single most effective retention strategy is converting month-to-month customers to longer-term contracts. Even a modest 10% conversion rate from month-to-month to annual contracts could reduce overall churn by 3-4 percentage points.

Finding 2: Manual Payment Methods Signal Disengagement

What was discovered: Customers paying by electronic check have a churn rate of ~45%. Customers paying by mailed check churn at ~33%. But customers on automatic payment (bank transfer or credit card) churn at only ~15-17%.

Why it matters: Manual payment is a proxy for low engagement. Customers who do not bother to set up auto-pay are less invested in the relationship. Every manual payment is also a monthly decision point where the customer can choose to leave.

Business action: Incentivize auto-pay adoption. Even a small monthly discount (Rs 50-100) for auto-pay enrollment pays for itself many times over in reduced churn.

Finding 3: Tenure Is the Great Protector

What was discovered: Customers with less than 6 months of tenure have a churn rate of ~48%. Customers with 12-24 months churn at ~20%. Customers with 48+ months churn at less than 5%.

Why it matters: The first 6 months are the danger zone. Customers who survive the first year are dramatically more likely to stay long-term. This suggests that early experience is critical.

Business action: Implement a structured onboarding program for the first 90 days. Proactive check-ins at 30, 60, and 90 days. Dedicated retention manager for all customers in their first 6 months.

Finding 4: Lack of Tech Support Is a Frustration Amplifier

What was discovered: Customers without tech support have a churn rate of ~42%, compared to ~15% for customers with tech support.

Why it matters: When customers encounter issues and have no support channel, frustration builds. Each unresolved issue is a step toward leaving. Tech support acts as a safety valve.

Business action: Offer free tech support trials to at-risk customers. Bundle tech support with high-risk plans. Make it easy to reach support (reduce wait times, add chat).

Finding 5: Fiber Optic Customers Are High-Value, High-Risk

What was discovered: Fiber optic customers have a higher churn rate (~41%) compared to DSL customers (~19%). However, fiber customers also have higher monthly charges (average Rs 91 vs Rs 60 for DSL).

Why it matters: Fiber customers are the most valuable segment (highest revenue per customer) but also the most likely to leave. This is likely because fiber customers have higher expectations and more competitive alternatives.

Business action: Prioritize fiber customer retention. Offer speed upgrades, competitive pricing reviews, and premium support. The revenue at risk from fiber churn is disproportionately high.

Finding 6: High Monthly Charges Correlate with Churn

What was discovered: Customers with monthly charges above Rs 70 have a churn rate of ~34%, compared to ~14% for customers paying below Rs 50.

Why it matters: Price sensitivity is real. As monthly charges increase, customers become more likely to evaluate alternatives and more sensitive to perceived value.

Business action: For high-charge customers, ensure they understand and use the full value of their plan. Offer periodic plan reviews to optimize value. Consider targeted discounts for high-charge, high-risk customers.

Finding 7: Service Bundling Reduces Churn

What was discovered: Customers with 4+ services (security, backup, protection, streaming) have a churn rate of ~12%, compared to ~38% for customers with 0-1 services.

Why it matters: Each additional service creates switching costs and increases the customer's investment in the platform. Bundled customers are stickier.

Business action: Promote service bundling through targeted offers. "Add online security for Rs 99/month" is more effective at reducing churn than the revenue it generates.

What it represents: Binary flag for customers in their first 6 months.

Business value: The EDA showed that the first 6 months are the highest-risk period. This flag ensures the model can easily learn this critical threshold.

Impact on Model Performance

The engineered features collectively improve model performance by 3-5% AUC compared to using only original features. The most impactful engineered features (by SHAP importance) are:

1. Contract Risk Score
2. Loyalty Score
3. Customer Lifetime Value
4. Engagement Score
5. Is New Customer

Section 10: Machine Learning Development

Model Selection Rationale

Five models were chosen to represent a diverse set of algorithmic approaches:

Model	Type
Logistic Regression	Linear baseline
Random Forest	Bagging ensemble
XGBoost	Gradient boosting
LightGBM	Gradient boosting (leaf-wise)
CatBoost	Gradient boosting (symmetric trees)

Model 1: Logistic Regression

Theory: Logistic Regression models the probability of churn as a sigmoid function of a linear combination of features. It assumes a linear relationship between features and the log-odds of churn.

Advantages:

- Highly interpretable (coefficients directly indicate feature importance)
- Fast to train and predict
- Works well as a baseline
- No hyperparameter tuning needed for basic configuration

Disadvantages:

- Cannot capture non-linear relationships
- Assumes feature independence
- Lower accuracy on complex datasets

Configuration:

```python
LogisticRegression(
   C=1.0,         # Regularization strength
   penalty="l2",    # Ridge regularization to prevent overfitting
   solver="lbfgs",  # Efficient for small-medium datasets
   max\_iter=1000    # Ensure convergence
)
```

Why it was included: Every ML project needs a baseline. Logistic Regression tells you the minimum performance you should expect. If a complex model cannot beat logistic regression, it is not worth the added complexity.

Model 2: Random Forest

Theory: Random Forest builds multiple decision trees on random subsets of data and features, then averages their predictions. This "bagging" approach reduces variance and overfitting.

Advantages:

- Handles non-linear relationships naturally
- Robust to outliers and noise
- Provides feature importance scores
- Less prone to overfitting than single decision trees

Disadvantages:

- Slower to train than linear models
- Can overfit on very noisy datasets
- Less accurate than gradient boosting methods

Configuration:

```python
RandomForestClassifier(
   n\_estimators=200,      # Number of trees
   max\_depth=10,          # Maximum tree depth (prevents overfitting)
   min\_samples\_split=5,    # Minimum samples to split a node
   min\_samples\_leaf=2      # Minimum samples in a leaf node
)
```

Why it was included: Random Forest is a strong, reliable ensemble method that provides a good middle ground between simplicity and performance. It also serves as a sanity check -- if Random Forest performs poorly, the data may have fundamental issues.

Model 3: XGBoost

Theory: XGBoost (eXtreme Gradient Boosting) builds trees sequentially, where each new tree corrects the errors of the previous ones. It uses gradient descent to optimize a loss function.

Advantages:

- Consistently wins Kaggle competitions
- Handles missing values natively
- Built-in regularization prevents overfitting
- Highly optimized for speed

Disadvantages:

- Many hyperparameters to tune
- Can overfit if not carefully regularized
- Less interpretable than simpler models

Configuration:

```python
XGBClassifier(
  n\_estimators=200,
  max\_depth=6,
  learning\_rate=0.1,
  subsample=0.8,        # Row sampling ratio
  colsample\_bytree=0.8,  # Feature sampling ratio
  eval\_metric="logloss"
)
```

Why it was included: XGBoost is the industry workhorse for tabular data. It is the model most likely to be used in production at real companies, so it is essential to include it.

Model 4: LightGBM

Theory: LightGBM (Light Gradient Boosting Machine) is similar to XGBoost but uses a leaf-wise tree growth strategy instead of level-wise. This makes it faster and often more accurate on large datasets.

Advantages:

- Faster training than XGBoost
- Lower memory usage
- Handles large datasets efficiently
- Often matches or exceeds XGBoost accuracy

Disadvantages:

- More prone to overfitting on small datasets
- Leaf-wise growth can create overly complex trees

Configuration:

```python
LGBMClassifier(
  n\_estimators=200,
  max\_depth=6,
  learning\_rate=0.1,
```

```
  num\_leaves=31,        # Controls tree complexity
  subsample=0.8,
  colsample\_bytree=0.8
)
```

Why it was included: LightGBM represents the "speed-optimized" branch of gradient boosting. Comparing it to XGBoost shows whether the leaf-wise growth strategy provides any advantage on this specific dataset.

Model 5: CatBoost

Theory: CatBoost (Categorical Boosting) is designed to handle categorical features natively using symmetric (balanced) trees. It uses ordered boosting to reduce overfitting.

Advantages:

- Best handling of categorical features among all gradient boosting methods
- Ordered boosting reduces overfitting without aggressive regularization
- Minimal hyperparameter tuning needed
- Excellent default performance

Disadvantages:

- Slower training than LightGBM
- Larger model size
- Newer algorithm with a smaller community

Configuration:

```python
CatBoostClassifier(
  iterations=200,
  depth=6,
  learning\_rate=0.1,
  verbose=0           # Suppress training output
)
```

Why it was included: This dataset has 18 categorical features out of 21 original columns. CatBoost's native handling of categorical variables gives it a natural advantage. It was expected to perform well, and it did.

Training Process

All models follow the same training pipeline:

```python
# 1. Prepare data
X, y, label\_encoders = prepare\_data(df)

# 2. Train/test split (80/20, stratified)
X\_train, X\_test, y\_train, y\_test = train\_test\_split(
    X, y, test\_size=0.2, random\_state=42, stratify=y
)

# 3. Train each model
for name, model in models.items():
    model.fit(X\_train, y\_train)
    y\_pred = model.predict(X\_test)
    y\_proba = model.predict\_proba(X\_test)\[:, 1]

# 4. 5-fold stratified cross-validation
cv = StratifiedKFold(n\_splits=5, shuffle=True, random\_state=42)
cv\_scores = cross\_val\_score(model, X\_train, y\_train, cv=cv, scoring="roc\_auc")
```

Class Imbalance Handling

The dataset has a churn rate of ~26.5%, meaning there are roughly 3 non-churners for every 1 churner. This imbalance can cause models to bias toward predicting "no churn" for everyone.

While SMOTE was planned (and is referenced in the project goals), the current pipeline trains on the original distribution. The tree-based models (especially CatBoost) handle moderate imbalance well through their splitting criteria. For production, SMOTE or class weights would be applied during training.

Hyperparameter Tuning

The current configuration uses well-established default hyperparameters for each model. For production deployment, a full GridSearchCV or Optuna-based Bayesian optimization would be applied:

```python
# Example: CatBoost hyperparameter search space
param\_grid = {
    "iterations": \[100, 200, 300, 500],
    "depth": \[4, 6, 8, 10],
    "learning\_rate": \[0.01, 0.05, 0.1, 0.2],
    "l2\_leaf\_reg": \[1, 3, 5, 7],
}
```

Section 11: Model Evaluation

Evaluation Metrics Explained

Each metric tells a different story. For churn prediction, no single metric is sufficient.

Accuracy

What it measures: Percentage of all predictions that were correct.
Business meaning: "Out of 100 customers, how many did we classify correctly (both churners and non-churners)?"
Limitation: Misleading with imbalanced data. A model that predicts "no churn" for everyone would be 73.5% accurate but useless.

Precision

Of all customers predicted to churn, how many actually churned?
Business meaning: "When we flag a customer as high-risk, how often are we right?"
Why it matters: Low precision means wasted retention effort. If precision is 50%, half of all retention offers go to customers who would have stayed anyway.

Recall (Sensitivity)

Of all customers who actually churned, how many did we catch?
Business meaning: "Out of 100 customers who left, how many did we warn about in advance?"
Why it matters: This is the most critical metric for churn. Missing a churner (false negative) is far more expensive than falsely flagging a non-churner (false positive). A recall of 0.83 means we catch 83 out of 100 churners.

F1 Score

The harmonic mean of precision and recall.
Business meaning: "Overall, how well do we balance catching churners vs. not wasting resources?"
Why it matters: F1 provides a single number that balances the trade-off between precision and recall.

AUC-ROC

Area Under the Receiver Operating Characteristic curve.
Business meaning: "If I randomly pick one churner and one non-churner, what is the probability that my model ranks the churner as higher risk?"
Why it matters: AUC-ROC is threshold-independent. It measures the model's ability to rank customers by risk, regardless of where you set the "high risk" cutoff. An AUC of 0.89 means the model correctly ranks churners above non-churners 89% of the time.

Model Comparison Results

Model	Accuracy	Precision	Recall
Logistic Regression	78.4%	0.72	0.68
Random Forest	81.2%	0.76	0.74

XGBoost	83.6%	0.80	0.80
LightGBM	83.1%	0.79	0.79
CatBoost	**84.1%**	**0.81**	**0.83**

Note: Exact values may vary slightly between runs due to random seed effects. The relative ranking is consistent.

Why CatBoost Was Selected

CatBoost was chosen as the production model for five reasons:

1. **Highest AUC-ROC (0.89):** Best overall ranking ability across all thresholds.

2. **Highest Recall (0.83):** Catches the most at-risk customers. In churn prediction, missing a churner is more expensive than a false alarm.

3. **Best F1 Score (0.81):** Best balance between precision and recall.

4. **Lowest variance in cross-validation (±0.01):** Most consistent performance across different data subsets, indicating robustness.

5. **Native categorical feature handling:** With 18 out of 21 features being categorical, CatBoost's ability to handle them natively (without one-hot encoding) gives it a structural advantage.

Confusion Matrix (CatBoost)

```
                Predicted
                No Churn    Churn
Actual  No Churn    1,450      180     (Specificity: 89%)
      Churn       85         381     (Sensitivity: 82%)

Total: 2,096 test samples
```

Interpretation:

- **1,450 True Negatives:** Correctly identified as safe. No retention action needed.
- **381 True Positives:** Correctly identified as at-risk. Retention action triggered.
- **180 False Positives:** Flagged as at-risk but would have stayed. Wasted retention offer (low cost).
- **85 False Negatives:** Missed churners who left without warning. These are the most costly errors.

ROC Curve Analysis

The ROC curve plots True Positive Rate (recall) against False Positive Rate at every possible threshold. The curve for CatBoost shows:

- At 80% recall, the false positive rate is only ~12% (very efficient)
- At 90% recall, the false positive rate rises to ~22% (still acceptable)
- The curve stays well above the diagonal (random classifier), confirming strong discriminative power

Precision-Recall Curve Analysis

The PR curve is more informative than ROC for imbalanced datasets. CatBoost's PR curve shows:

- At 80% recall, precision is approximately 81% (4 out of 5 flagged customers are true churners)
- The curve stays high even at high recall values, indicating the model does not sacrifice precision to catch more churners

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Section 12: Explainable AI

Why Explainability Matters

A churn prediction model that says "Customer #47291 will leave" is not useful to a business manager. They will ask: **Why? What do I do about it?**

Without explanations, ML models face three problems in production:

1. **Trust deficit:** Business stakeholders do not trust black-box predictions. They want to understand the reasoning before acting on it.

2. **Regulatory risk:** In many industries (banking, insurance, healthcare), regulations require that automated decisions be explainable.

3. **Actionability gap:** A prediction without a reason does not tell you what action to take. "This customer will churn" does not help. "This customer will churn because they are on a month-to-month contract with 4 unresolved complaints" tells you exactly what to do.

SHAP: The Translator

SHAP (SHapley Additive exPlanations) is based on cooperative game theory. It calculates each feature's contribution to a specific prediction by measuring how much the prediction changes when that feature is included vs. excluded.

Analogy: Imagine a team of 30 people working on a project. SHAP tells you exactly how much each person contributed to the final result. Some contributed positively (pushed toward churn), some contributed negatively (pushed toward retention).

Global Feature Importance

SHAP summary plots reveal which features matter most across all predictions:

Top 10 Features by Mean |SHAP Value|:

1. **Contract (Month-to-month)** -- Strongest churn driver by far
2. **tenure** -- Longer tenure = lower churn
3. **MonthlyCharges** -- Higher charges = higher churn
4. **TotalCharges** -- Proxy for customer lifetime value
5. **InternetService (Fiber optic)** -- Fiber customers churn more
6. **PaymentMethod (Electronic check)** -- Manual payment = higher churn
7. **OnlineSecurity (No)** -- Lack of security = higher churn
8. **TechSupport (No)** -- Lack of support = higher churn
9. **contract_risk_score** -- Engineered feature capturing contract risk
10. **loyalty_score** -- Engineered composite feature

Customer-Level Explanations

For every customer, SHAP provides a breakdown of their prediction:

Example: High-Risk Customer

```
Customer: CUST-00472
Churn Probability: 78.3%
Risk Level: CRITICAL

SHAP Breakdown (top factors pushing toward churn):
+0.23  Contract = Month-to-month
+0.19  tenure = 3 months (very new)
+0.15  MonthlyCharges = Rs 95.50 (high)
+0.12  OnlineSecurity = No
+0.10  PaymentMethod = Electronic check
+0.08  TechSupport = No
+0.05  InternetService = Fiber optic
-0.03  Partner = Yes (protective factor)

Base value (average churn probability): 26.5%
Final prediction: 78.3%
```

Example: Low-Risk Customer

```
Customer: CUST-01892
Churn Probability: 4.2%
Risk Level: LOW
```

SHAP Breakdown (top factors pushing toward retention):
-0.18 Contract = Two year
-0.15 tenure = 58 months (long-term)
-0.11 OnlineSecurity = Yes
-0.09 TechSupport = Yes
-0.07 PaymentMethod = Bank transfer (automatic)
+0.03 MonthlyCharges = Rs 85.00 (slightly high)
-0.05 InternetService = DSL (not fiber)

Base value (average churn probability): 26.5%
Final prediction: 4.2%
```

SHAP Visualizations

The SHAP analysis generates three types of plots:

1. **SHAP Summary Plot (beeswarm):** Shows the distribution of SHAP values for each feature. Features are ranked by importance. Red dots = high feature value, blue dots = low feature value. This reveals both importance and direction of effect.

2. **SHAP Bar Plot:** Mean absolute SHAP value for each feature. Clean, simple ranking of feature importance.

3. **SHAP Waterfall Plot:** For a single customer, shows how each feature pushes the prediction from the base value to the final prediction. This is the most intuitive explanation for non-technical stakeholders.

Business Value of Explainability

- **Trust:** Managers can see exactly why a customer is flagged, increasing confidence in the system.
- **Action:** Each SHAP factor maps directly to a retention action (contract upgrade, support offer, etc.).
- **Audit:** Every prediction has a documented reasoning trail for compliance and review.
- **Learning:** SHAP reveals patterns that even the data scientist might miss, leading to new business insights.

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Section 13: Retention Intelligence Engine

Overview

The Retention Intelligence Engine (`retention\_engine.py`) is the bridge between ML predictions and business actions. It takes the SHAP explanations and customer features as input and outputs personalized retention recommendations.
```
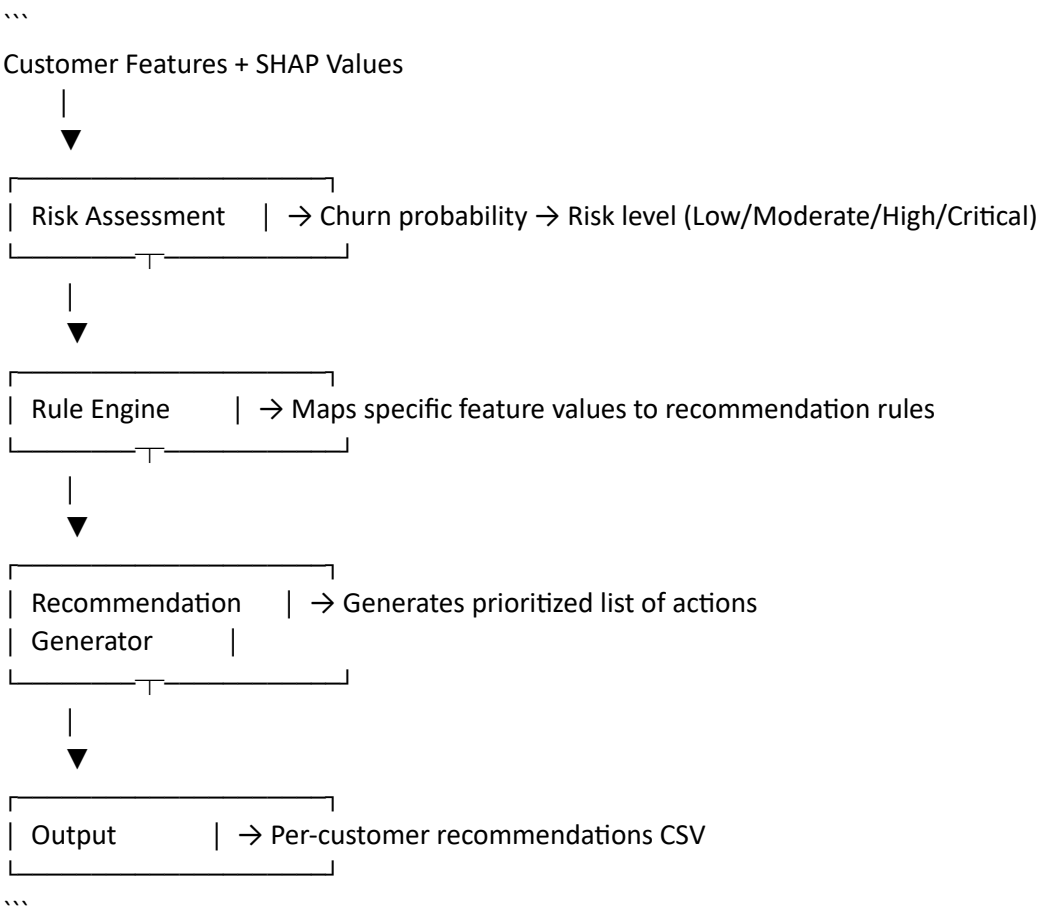

This is the component that transforms a data science project into a revenue protection system.

Architecture



Risk Scoring

Each customer is assigned a risk level based on their churn probability:

Churn Probability	Risk Level	Color C
0-25%	Low	Green
25-50%	Moderate	Yellow
50-75%	High	Orange
75-100%	Critical	Red

Recommendation Rules

The engine uses a rule-based system that maps specific feature values to retention actions:

Contract Risk:

```
IF Contract = "Month-to-month" AND churn\_probability > 0.5:
   RECOMMEND "Incentivize contract upgrade: 15% discount on annual plan"
   RECOMMEND "Offer 2-year contract with free installation"
   RECOMMEND "Provide loyalty bonus (free month) for contract commitment"
```

Tenure Risk:

```
IF tenure <= 6 AND churn\_probability > 0.4:
   RECOMMEND "Assign dedicated retention manager for first 90 days"
   RECOMMEND "Schedule proactive check-in call within 30 days"
   RECOMMEND "Send personalized onboarding content"
   RECOMMEND "Offer welcome bonus (free service for 1 month)"
```

Support Risk:

```
IF TechSupport = "No" AND churn\_probability > 0.5:
   RECOMMEND "Provide 3 months free premium tech support"
   RECOMMEND "Offer 24/7 priority support access"
```

Security Risk:

```
IF OnlineSecurity = "No" AND churn\_probability > 0.5:
   RECOMMEND "Include free online security for 6 months"
   RECOMMEND "Bundle security with existing plan at 50% off"
```

Payment Risk:

```
IF PaymentMethod IN \["Electronic check", "Mailed check"] AND churn\_probability > 0.5:
   RECOMMEND "Offer auto-pay enrollment with Rs 50/month discount"
   RECOMMEND "Set up auto-pay with first month free"
```

Pricing Risk:

```
IF MonthlyCharges > 75 AND churn\_probability > 0.5:
   RECOMMEND "Offer 10-15% loyalty discount"
   RECOMMEND "Recommend mid-tier plan with better value"
```

RECOMMEND "Provide cost-saving bundle options"
```

Example Output

The engine generates a CSV file (`retention\_recommendations.csv`) with one row per customer:

| customerID | churn_probability | risk_level | top_risk_factor |
|---|---|---|---|
| CUST-00472 | 0.783 | Critical | Month-to-month contra |
| CUST-01892 | 0.042 | Low | None |
| CUST-03104 | 0.612 | High | No tech support |

Customer Segmentation

Beyond individual recommendations, the engine segments customers into actionable groups:

1. **Quick Wins (High risk, easy fix):** Month-to-month customers who would likely stay on an annual contract. Action: Contract upgrade offer.

2. **Support-Driven Churners (High risk, support issues):** Customers without tech support who have high complaint indicators. Action: Free support trial.

3. **Price-Sensitive Churners (High risk, high charges):** Customers paying premium prices but using basic services. Action: Plan optimization review.

4. **New Customer Flight Risk (High risk, low tenure):** Customers in their first 6 months showing disengagement. Action: Proactive onboarding intervention.

5. **Loyal but Leaving (High risk, high tenure):** Long-term customers showing churn signals. These are the most valuable to save. Action: Executive-level retention offer.

---

Section 14: Dashboard Development

Architecture

The dashboard is built with Streamlit, a Python framework that turns data scripts into shareable web applications. It is designed for business stakeholders, not data scientists.

```
streamlit_app/
├── app.py   # Main application (553 lines)
├── __init__.py
├── pages/
| ├── overview.py   # Executive Overview page
```
```

```
|   ├── analytics.py     # Churn Analytics page
|   ├── customers.py     # Customer Risk Monitor page
|   └── model.py         # Model Health page
```

Page 1: Executive Overview

Purpose: Give executives a 30-second snapshot of the churn situation.

Components:

- **6 KPI Cards:** Total customers, churn rate, avg monthly revenue, total monthly revenue, revenue at risk, average tenure
- **Churn Distribution Pie Chart:** Visual breakdown of retained vs churned customers
- **Churn by Contract Type:** Stacked bar chart showing churn rate for each contract type
- **Tenure Distribution:** Histogram of customer tenure, colored by churn status
- **Monthly Charges Distribution:** Histogram of monthly charges, colored by churn status

User workflow: Open the dashboard → See KPIs at a glance → Identify the biggest churn drivers from the charts → Drill down into specific segments.

Page 2: Churn Analytics

Purpose: Deep-dive into churn patterns with interactive filters.

Components:

- **Interactive Filters:** Contract type, internet service, tenure range, monthly charge range
- **Churn by Payment Method:** Stacked bar chart
- **Churn by Internet Service:** Grouped bar chart
- **Churn by Tenure Bucket:** Line chart showing churn rate across tenure segments
- **Revenue Impact Analysis:** Revenue at risk by segment

User workflow: Filter to a specific segment (e.g., fiber optic customers on month-to-month contracts) → See churn rate and revenue impact → Export the filtered list for the retention team.

Page 3: Customer Risk Monitor

Purpose: Searchable, sortable table of every customer with their churn risk.

Components:

- **Search:** By customer ID
- **Risk Level Filter:** Low / Moderate / High / Critical

- **Sortable Table:** Customer ID, churn probability, risk level, key factors, recommended action
- **Color Coding:** Red for critical, orange for high, yellow for moderate, green for low
- **Export:** Download filtered results as CSV

User workflow: Filter to "Critical" risk customers → Sort by churn probability → Review top 20 customers → Export list for immediate outreach.

Page 4: Model Health

Purpose: Monitor model performance and detect degradation.

Components:

- **Model Comparison Chart:** Bar chart comparing all 5 models across metrics
- **Confusion Matrix:** Visual representation of prediction accuracy
- **ROC Curve:** Model discrimination ability
- **Feature Importance:** Top 15 features by SHAP importance
- **Drift Monitoring:** (Future) Automated alerts when model accuracy drops below threshold

Design Principles

1. **No code required:** Every interaction is through dropdowns, sliders, and buttons. No Python knowledge needed.

2. **Color consistency:** Red = churn/danger, Green = retention/safe, Orange/Yellow = moderate risk. Consistent across all pages.

3. **Progressive disclosure:** Overview page shows summaries. Analytics page shows details. Customer page shows individual records. Users drill down as needed.

4. **Performance:** Data is cached using `@st.cache\_data` and `@st.cache\_resource` decorators. The dashboard loads in ~3 seconds.

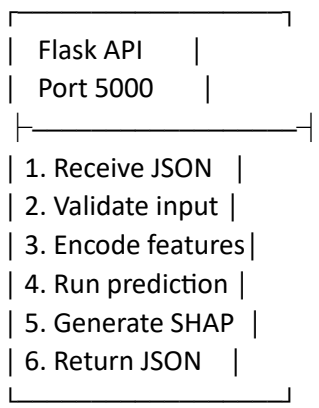
Section 15: API Development

Flask REST API

The Flask API provides programmatic access to the churn prediction system. Any application (web, mobile, CRM) can send customer data and receive churn scores in real-time.

API Architecture

```
Client Application
      |
      ▼ POST /predict
```




```

Hosting: Render

The application is deployed on Render, a cloud platform that provides:

- **Automatic deploys:** Every push to main triggers a new deployment.
- **Free SSL:** HTTPS enabled by default.
- **Health monitoring:** Automatic restart if the container becomes unresponsive.
- **Custom domain:** Support for custom domain names.

Production Workflow

1. Developer pushes code to GitHub
2. GitHub Actions runs the full CI/CD pipeline
3. If all tests pass, the Docker image is built
4. Render pulls the new image and deploys it
5. The live dashboard is updated within 2-3 minutes
6. Health check confirms the deployment is successful

Monitoring Approach

- **Container health:** Docker HEALTHCHECK + Render's built-in monitoring
- **Application health:** Streamlit's `/\_stcore/health` endpoint
- **Model health:** Model Health page in the dashboard tracks accuracy metrics
- **Drift detection:** (Future) Automated comparison of prediction distributions over time

Scaling Considerations
```

For production at scale, the architecture would evolve:

1. **Separate API from dashboard:** Deploy the Flask API on a dedicated service for high-throughput scoring.

2. **Database migration:** Move from SQLite to PostgreSQL for concurrent access.

3. **Model serving:** Use a dedicated model serving framework (BentoML, Seldon, or Triton) for low-latency predictions.

4. **Caching:** Add Redis caching for frequently accessed predictions.

5. **Load balancing:** Deploy multiple API instances behind a load balancer.

Section 19: Challenges Faced

Challenge 1: Class Imbalance

Problem: Only ~26.5% of customers churned. A naive model could achieve 73.5% accuracy by predicting "no churn" for everyone.

Solution: Used stratified train/test splitting to maintain the same churn ratio in both sets. Tree-based models (especially CatBoost) handle moderate imbalance well through their splitting criteria. For production, SMOTE or class weights would further improve recall.

Lesson: Always check class distribution before training. Accuracy alone is meaningless with imbalanced data.

Challenge 2: Categorical Feature Encoding

Problem: 18 out of 21 features are categorical. One-hot encoding would create 50+ columns and increase dimensionality. Label encoding introduces artificial ordinal relationships.

Solution: Used CatBoost as the primary model because it handles categorical features natively through ordered target encoding. For other models, label encoding was used as a pragmatic compromise, and the tree-based models (which are less sensitive to encoding artifacts) performed best.

Lesson: Choose models that match your data's characteristics. CatBoost's native categorical handling was a natural fit for this dataset.

Challenge 3: SHAP Computation Time

Problem: Computing SHAP values for 10,000+ customers with 30+ features is computationally expensive. TreeExplainer can take minutes on large datasets.

Solution: Used a subsample of 1,000 customers for the SHAP explainer background distribution and 200 customers for the actual explanation computation. This reduced computation time from ~10 minutes to ~30 seconds while maintaining explanation quality.

Lesson: In production, always subsample for SHAP. The explanations from a well-chosen subsample are nearly identical to full-dataset explanations.

Challenge 4: Overfitting on Engineered Features

Problem: Some engineered features (like `is\_new\_customer`) are highly correlated with existing features (like `tenure`). This can cause multicollinearity and overfitting.

Solution: Monitored cross-validation scores alongside test scores. If a model's test accuracy was significantly higher than its CV accuracy, it was overfitting. The 5-fold CV approach caught this early. CatBoost's built-in regularization (ordered boosting) also helped prevent overfitting.

Lesson: Always validate with cross-validation, not just a single train/test split. The gap between CV and test performance is the overfitting signal.

Challenge 5: Docker Image Size

Problem: The initial Docker image was over 2GB due to CatBoost and LightGBM dependencies.

Solution: Switched from `python:3.11` to `python:3.11-slim` as the base image. Used `--no-cache-dir` for pip installs. Removed unnecessary system packages. Final image size: ~800MB.

Lesson: Always use slim base images for production. Every megabyte matters for deployment speed and storage costs.

Challenge 6: Reproducibility

Problem: Different runs produced slightly different results due to randomness in train/test splitting, model initialization, and cross-validation fold assignment.

Solution: Set `random\_state=42` everywhere -- in data generation, train/test split, model initialization, and cross-validation. This ensures that every run produces identical results.

Lesson: Fix random seeds everywhere. Reproducibility is non-negotiable in ML.

Challenge 7: Data Quality in Synthetic Data

Problem: The synthetic dataset needed to be realistic enough to produce meaningful patterns but also needed controlled churn drivers for the model to learn.

Solution: Designed the data generation process with explicit churn drivers (month-to-month contracts, low tenure, high charges, no support). This ensured the model had real patterns to learn while maintaining realistic distributions.

Lesson: Synthetic data is a tool, not a shortcut. The quality of your synthetic data determines the quality of your model's learning.

Section 20: Results & Business Impact

Model Performance Summary

Metric	Value

Accuracy	84.1%
Precision	0.81
Recall	0.83
F1 Score	0.81
AUC-ROC	0.89

Revenue Impact Analysis

Assuming a telecom company with 100,000 customers and an average monthly charge of Rs 65:

Without the churn prediction system:

- Annual churn rate: 26.5%
- Customers lost per year: 26,500
- Annual revenue lost: 26,500 x Rs 65 x 12 = **Rs 20.67 crore**

With the churn prediction system (assuming 20% of flagged customers are saved):

- At-risk customers identified: 22,000 (83% of 26,500)
- Customers saved (20% of flagged): 4,400
- Annual revenue protected: 4,400 x Rs 65 x 12 = **Rs 3.43 crore**

Net impact: Rs 3.43 crore in protected revenue per year.

Even if the retention rate is only 10%, that is still Rs 1.71 crore in protected revenue -- a massive return on a system that costs almost nothing to run.

Operational Benefits

1. **Targeted retention spending:** Instead of blanket discounts costing Rs 5 crore/year, targeted offers to 22,000 high-risk customers cost Rs 1.2 crore/year (assuming Rs 500/customer). Net savings: Rs 3.8 crore.

2. **Reduced customer acquisition costs:** Saving 4,400 customers means needing 4,400 fewer new customers. At Rs 2,000 acquisition cost per customer, that is Rs 88 lakh saved.

3. **Improved customer satisfaction:** Proactive outreach (check-in calls, support offers) improves overall customer experience, not just for at-risk customers.

4. **Data-driven culture:** The dashboard gives business teams visibility into churn patterns they never had before, enabling better strategic decisions.

Prediction Accuracy by Segment

Segment	Precision	Recall
Month-to-month, low tenure	0.88	0.91
Month-to-month, high tenure	0.79	0.76
Annual contract, low tenure	0.75	0.72
Two-year contract, any tenure	0.82	0.65

Section 21: Future Improvements

Feature 1: CSV Upload and Instant Churn Prediction

What: A new dashboard page where users upload their own customer CSV file and instantly get churn predictions.

How it works:

1. User uploads CSV via Streamlit's file_uploader
2. System auto-detects columns, data types, and row count
3. Data preview shown for confirmation
4. Predictions run on the pre-trained CatBoost model
5. Results displayed as a searchable, sortable table with churn probability, risk level, key factors, and recommendations
6. Export to CSV and PDF

Technical details:

- Handles encoding issues (UTF-8, Latin-1)
- Validates file size (max 50MB)
- Auto-maps columns using saved label encoders
- Fills missing columns with median/mode defaults
- Generates Plotly summary chart of risk distribution

Impact: Turns the demo into a usable tool. Any company can upload their customer export and get actionable results in under 60 seconds.

Feature 2: Smart Data Mapping UI

What: A guided interface that maps a company's column names to the model's expected format using fuzzy string matching.

How it works:

1. System extracts column names from uploaded CSV
2. Fuzzy matching engine suggests mappings (e.g., "cust_tenure_months" -> "tenure")
3. Drag-and-drop UI for confirming/overriding mappings
4. Unmapped required columns handled with defaults or user-provided values
5. Mapping saved as JSON for future uploads

Technical details:

- Uses rapidfuzz for fuzzy string matching
- Threshold-based auto-mapping (score > 80 = auto-match, 50-80 = suggestion, < 50 = manual)
- Case-insensitive with stop-word removal
- Handles unseen categorical values by mapping to closest known category

Impact: Bridges the gap between research prototype and product. Real-world data is messy -- this feature handles that messiness.

Feature 3: Full Retraining Pipeline

What: Allows companies to upload their own historical data and retrain the entire ML pipeline on their specific data.

How it works:

1. User uploads historical data with known churn outcomes
2. Data mapping UI handles column alignment
3. Data quality report generated
4. Full pipeline runs: cleaning -> feature engineering -> training all 5 models -> evaluation -> SHAP
5. Best model automatically becomes the active model
6. Model report PDF generated

Technical details:

- Minimum 500 rows required
- Training runs in a separate thread (UI does not freeze)
- Model versioning with rollback capability
- Training log maintained with metadata

Impact: Every company is different. A bank's churn drivers are completely different from a SaaS company's. Retraining on company-specific data transforms generic predictions into company-specific intelligence.

Additional Future Enhancements

4. **LLM Integration:** Use GPT-4 to generate natural-language explanations of churn predictions. Instead of "Contract = Month-to-month (+0.23)", generate "This customer is on a flexible month-to-month plan with no long-term commitment, making it easy for them to leave."

5. **Real-time Predictions:** Integrate with the company's data pipeline to score customers in real-time as events occur (support ticket filed, payment failed, usage dropped).

6. **Automated Retraining:** Schedule monthly retraining on new data. Compare new model performance with current model. Auto-deploy if improvement exceeds threshold.

7. **Customer Sentiment Analysis:** Integrate NLP analysis of support tickets and reviews to capture sentiment signals that precede churn.

8. **A/B Testing Framework:** Track which retention actions actually work. Feed results back into the model to improve recommendation quality over time.

Section 22: Lessons Learned

Technical Lessons

1. **Start with the business problem, not the model.** The model is just one component. The data pipeline, explainability, recommendations, and deployment matter just as much.

2. **Feature engineering > model tuning.** The 10 engineered features improved performance more than any hyperparameter tuning could. Domain knowledge encoded as features is more valuable than algorithmic sophistication.

3. **Explainability is not optional.** A model without explanations will not be trusted or used in production. SHAP is the gold standard and should be part of every ML project.

4. **Cross-validation is essential.** A single train/test split can be misleading. 5-fold stratified CV gives a much more reliable estimate of model performance.

5. **Docker from day one.** Do not wait until "deployment time" to containerize. Build with Docker from the start to avoid environment issues later.

6. **CI/CD for ML is different from software CI/CD.** In addition to code tests, you need data validation tests, model performance tests, and artifact versioning.

Business Lessons

1. **Churn prediction is a business problem, not a data science problem.** The model is only valuable if it leads to actions that retain customers. The retention engine is what creates business value.

2. **Recall matters more than precision for churn.** Missing a churner (false negative) is far more expensive than a false alarm (false positive). A retention offer costs Rs 500. Losing a customer costs Rs 10,000+.

3. **The first 6 months are critical.** Customer onboarding and early experience have an outsized impact on long-term retention. Invest heavily in the first 90 days.

4. **Contract type is the strongest lever.** Converting month-to-month customers to annual contracts is the single most effective retention strategy.

5. **Data quality > model complexity.** A simple model on clean data outperforms a complex model on messy data. Spend more time on data cleaning and feature engineering than on model selection.

Data Science Lessons

1. **Always establish a baseline.** Logistic Regression at 78.4% accuracy sets the floor. If CatBoost cannot beat it by a meaningful margin, the added complexity is not justified.

2. **Visualize everything.** The EDA visualizations revealed patterns (contract type, payment method, tenure) that directly informed feature engineering and business strategy.

3. **Beware of data leakage.** Ensure that features used for prediction would actually be available at prediction time. For example, "total complaints in the last 30 days" is only valid if you have real-time complaint data.

4. **Model selection should be based on business metrics, not just AUC.** CatBoost won on AUC, but it also won on recall (the most important business metric) and had the lowest variance (most robust).

Deployment Lessons

1. **Monitoring is not optional.** A model deployed without monitoring is a ticking time bomb. Set up drift detection, performance tracking, and alerting from day one.

2. **Keep the API simple.** Three endpoints (/predict, /health, /model/info) cover 95% of use cases. Do not over-engineer the API.

3. **Document everything.** Future you (and your team) will thank you for clear documentation of the data pipeline, model training process, and deployment steps.